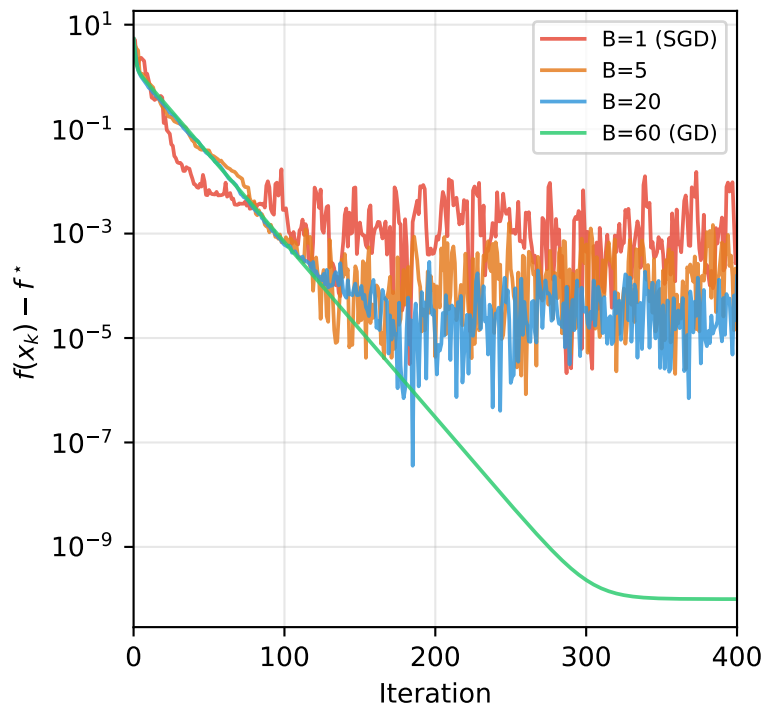
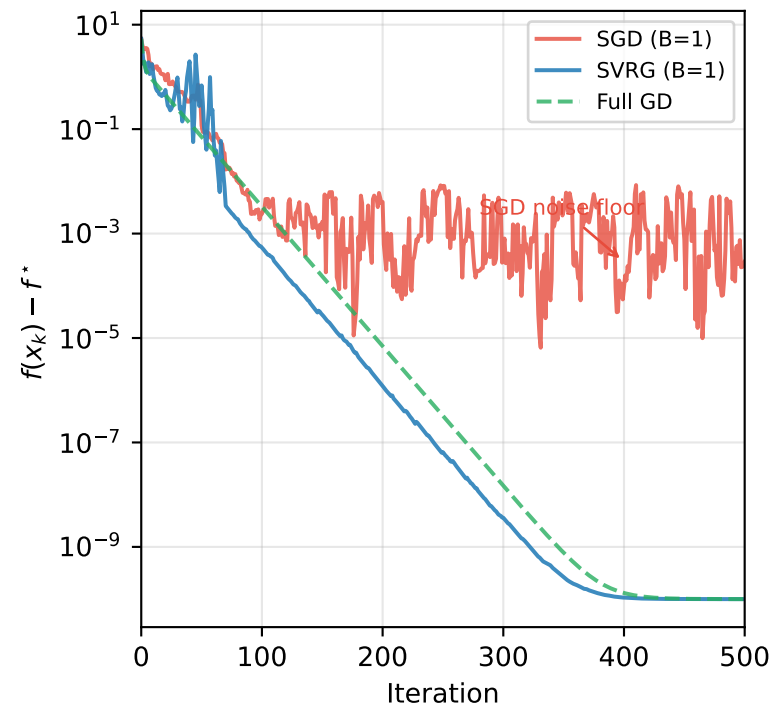


Stochastic Gradient Descent: Variance, Noise, and Convergence Theory

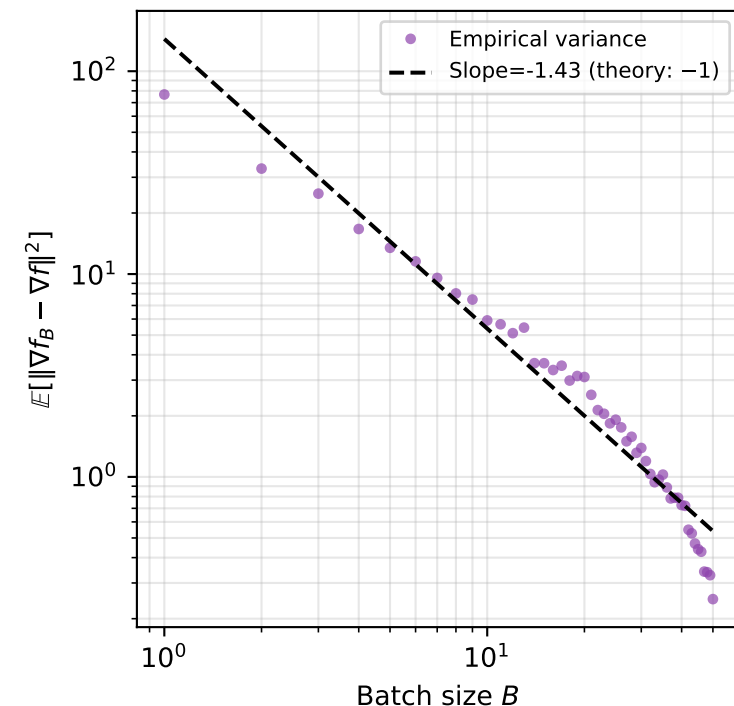
SGD: Batch Size Effect



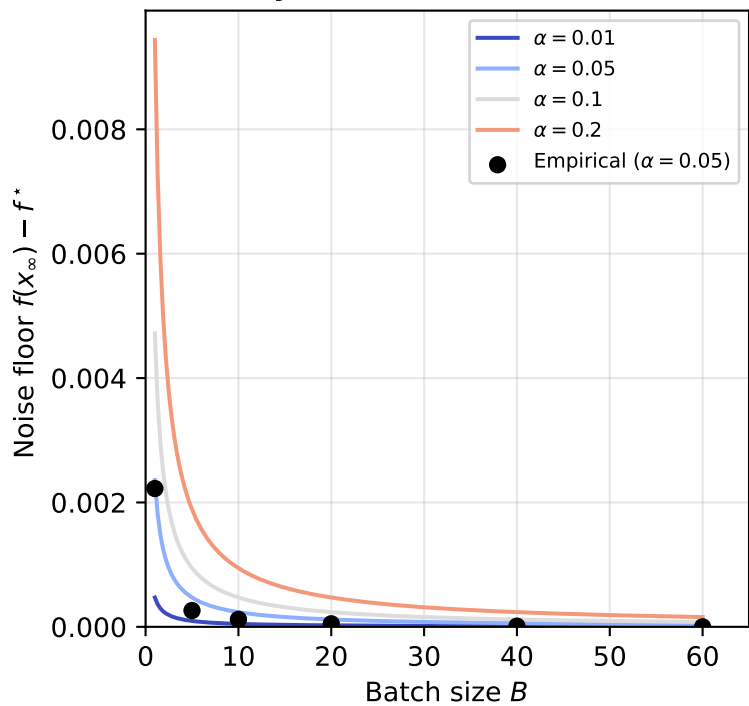
SGD vs SVRG



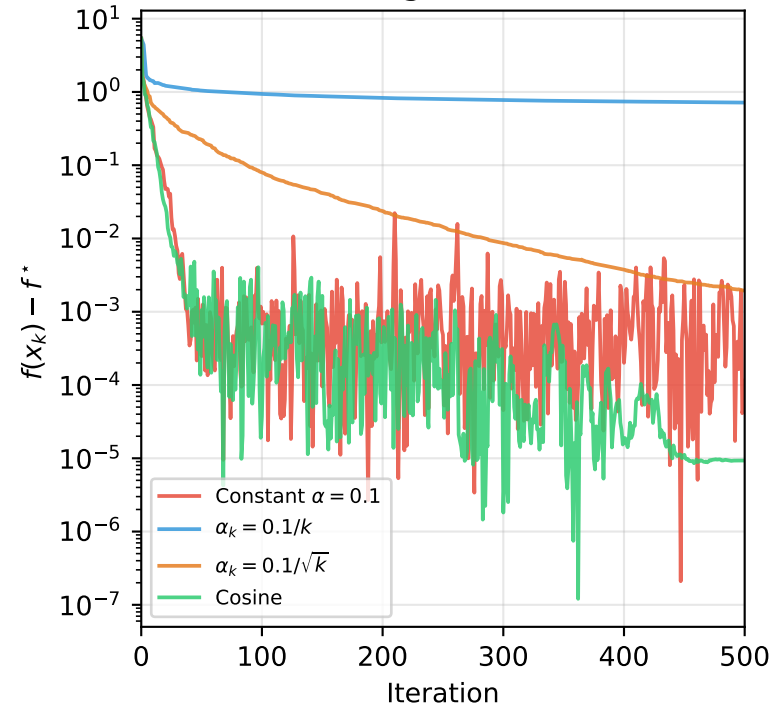
Gradient Variance $\propto 1/B$



Steady-State Error vs Batch Size



SGD Learning Rate Schedules



SGD Convergence Theory

